📘 MODULE 12: SUBSEARCHES

12.1 Concept

A **subsearch** is a search inside square brackets [ ]. It runs **first**, and its results are fed into the outer search as filter conditions.

12.2 Why it matters

“Show all web activity from the IPs that had many failed logins.” One search finds the IPs, and the other uses them.

12.3 How it works (the key to understanding it)

spl

index=web [search index=auth "Failed password" | stats count by src_ip | where count>10 | fields src_ip]

1. Splunk runs the inner search and gets: src_ip = 1.1.1.1, 2.2.2.2.
2. It rewrites the outer search as:

index=web ((src_ip="1.1.1.1") OR (src_ip="2.2.2.2"))

3. Then it runs the outer search.

Preview what a subsearch returns by running the inner search followed by | format. You’ll see the generated OR string.

12.4 Rules

- The inner search **should end with fields <name>** so the count column doesn’t also get added as a filter.
- The returned **field names must match the outer search’s field names**. Use rename inside the subsearch if they differ.
- Return a single value with return: [search … | return 10 src_ip] (top 10 values of src_ip).
- Subsearches starting with a generating command need a leading pipe: [| inputlookup bad_ips.csv | fields ip].

12.5 Limits (very common interview question)

Default limits (limits.conf, [subsearch] stanza): about **10,000 results** and about **60 seconds** runtime. If exceeded, results are **silently truncated**, so your outer search may be wrong with no error. Verify the values for your version in the docs.

12.6 Worked examples

Example 1: filter by a subsearch

spl

```
index=_internal sourcetype=splunkd_access
[ search index=_internal sourcetype=splunkd_access status>=400
  | stats count by clientip
  | where count>5
  | fields clientip ]
| stats count by clientip, status
```

What it does: finds IPs with more than 5 error responses, then shows all their traffic by status.
Common mistake: leaving count in the subsearch output, which adds count=... as a filter and returns nothing.

Example 2: subsearch from a lookup

spl

```
index=* sourcetype=linux_secure
[ | inputlookup threat_ips.csv | rename ip AS src_ip | fields src_ip ]
```

12.7 Alternatives (usually better)

Instead of	Use
Subsearch for a static list	A **lookup** with lookup + where
Subsearch to compare against a total	**eventstats** (Module 14)
Subsearch to combine two datasets	One search with **stats** on both

12.8 Hands-on

1. Find the host with the most ERROR events in splunkd, then show all its events (subsearch with head 1).
2. Use | format on your subsearch to see the generated filter.
3. Rewrite a subsearch you wrote as a lookup or stats version and compare speed.

12.9 Troubleshooting

Symptom	Cause
Outer search returns nothing	Field name mismatch, or count field returned
Results look incomplete	Hit 10,000 / 60-second limit
Error about command	Missing leading | or missing search keyword
Slow search	Subsearch is heavy, so consider restructuring

12.10 Interview Q&A

- **What is a subsearch?** A bracketed search whose results become filter conditions for the outer search.
- **What are its limits?** Default 10,000 results and 60 seconds, with silent truncation.
- **Which runs first?** The subsearch.
- **Why must field names match?** Because the results become field=value filters on the outer search.
- **Alternatives?** Lookup, stats over combined data, eventstats.

12.11 Revision

[ ] runs first · returns field=value OR … · end with fields · rename to match · limits ~10k / 60s · use format to debug.